

- What is the difference between logical design, physical design, and signoff?
- What information is contained in a technology LEF?
- What is the difference between LEF and DEF?
- What is the purpose of a Liberty .lib file?
- What information does a standard-cell .lib contain?
- What is an SDC file, and why is it important in Physical Design?
- What are the main steps involved in floorplanning?
- What factors should be considered when deciding the core utilization?
- What is core utilization, and why should utilization not be made too high?
- What is the difference between core area, die area, and standard-cell area?
- What is aspect ratio in floorplanning?
- What is a floorplan blockage?
- What is the difference between placement blockage, routing blockage, and keepout margin?
- What are macros, and why are macro locations important?
- What factors do you consider when deciding macro placement?
- Why are macros generally placed near the boundary of the core?
- What problems can occur if macros are placed too close together?
- What is a macro channel, and why is sufficient routing space required between macros?
- What is halos around macros, and why are they used?
- What are power rings, power stripes, and power rails?
- Why are multiple power straps used across a large chip?
- What is IR drop?
- What is the difference between static IR drop and dynamic IR drop?
- How can you fix an electromigration violation?
- What is the difference between IR drop and electromigration?
- What is standard-cell placement?
- What is congestion?
- What are the different types of routing congestion?
- How can you identify whether congestion is caused by placement or routing resources?
- What are common techniques to reduce congestion?
- How can macro placement affect congestion?
- What is placement optimization?
- What is timing-driven placement?
- What is the difference between wire delay and cell delay?
- Why does wire delay become increasingly important in advanced technology nodes?
- What is the difference between positive skew and negative skew?
- What is the difference between source latency and network latency?



- What is the difference between setup timing and hold timing?
- Why is setup checked at one clock edge and hold at another?
- Why is cell upsizing useful for setup optimization?
- Why can upsizing a cell increase power?
- How does buffer insertion help timing?
- When would you use cell downsizing?
- What is the difference between data-path optimization and clock-path optimization?
- What is the relationship between clock period and setup timing?
- What is the difference between crosstalk delay and crosstalk noise?
- What is the difference between signal routing, clock routing, and power routing?
- What are routing layers?
- Why are higher metal layers generally preferred for long-distance/global connections?
- What is preferred routing direction?
- Why do adjacent metal layers usually have different preferred directions?
- Why does the antenna effect occur during fabrication?
- How can antenna violations be fixed?
- What is a diode-based antenna fix?
- What is metal jumping, and how can it help with antenna violations?
- What is metal density?
- Why is metal density important for manufacturing?
- What is CMP, and how is it related to metal density?
- What are decap cells? Why are decap cells inserted?
- What is the difference between buffering and upsizing as timing fixes?
- What is a high-fanout net?
- Why are reset and enable signals often high-fanout nets?
- How would you optimize a high-fanout reset net?
- What is congestion-driven optimization?
- If you have a setup violation after routing, what would you check first?
- If you have a hold violation after CTS, what possible fixes would you consider?
- If congestion is extremely high in one region, how would you debug it?
- If a macro is causing severe routing congestion, what physical changes could you make?
- If IR drop is high in one region, how would you debug and fix it?
- If an EM violation occurs on a power strap, what solutions could you consider?
- If timing is clean before routing but fails badly after routing, why could this happen?
- If setup timing is clean but hold timing becomes very bad after CTS, what could be the reason?
- If a design has excellent timing but very high congestion, what trade-offs would you consider?



- If utilization is reduced and congestion improves but timing becomes worse, why?
- If increasing the clock frequency causes setup violations, what parts of the design would you investigate?
- If one timing path has very large net delay compared with cell delay, what does that indicate?
- If a critical net has large capacitance, what physical changes could improve its timing?
- If a critical path has high fanout, how would you optimize it?
- If you are given a block with high utilization, high congestion, setup violations, and IR-drop violations, in what order would you approach the problems and why?
- Given a design that fails timing after route, explain your complete timing-closure debug methodology.
- Given a design that fails both setup and hold timing, explain how you would prioritize and fix the violations without creating new problems.
- Given a design with severe congestion around macros, explain how you would analyze the root cause and modify the floorplan.
- You receive a design with 85% utilization, multiple macros, several clock domains, high congestion, setup violations, and IR-drop violations. Explain how you would approach the Physical Design flow from floorplanning through signoff.
- Your design has 85% utilization and severe routing congestion after placement. How would you identify the root cause and what changes would you try?
- You reduce utilization from 85% to 70%, but congestion improves while timing becomes worse. Why could this happen?
- A macro is causing severe congestion around its boundaries. What would you check, and how would you modify the floorplan?
- Two large macros are placed very close to each other and routing congestion is extremely high between them. What would you do?
- You have sufficient core area, but routing congestion is still very high. What could be the possible reasons?
- Congestion is concentrated only in one region of the design. How would you debug it?
- After placement, one timing path has extremely high net delay while cell delay is relatively small. What does this indicate, and how would you fix it?
- A critical path has very high fanout. What physical optimization techniques would you consider?
- A critical net has very large capacitance. What could be causing it, and how would you reduce it?
- Your design passes timing after placement but develops significant setup violations after CTS. What could have caused this?



- Your design has good setup timing before CTS but severe hold violations immediately after CTS. What would you investigate?
- After CTS, the clock skew is extremely high. What could be the reasons, and how would you fix it?
- One clock branch has significantly larger insertion delay than the other branches. How would you debug it?
- You have negative skew on a critical path. How does it affect setup and hold timing?
- CTS improves setup timing but creates thousands of hold violations. Why can this happen?
- You fix hold violations by adding buffers, but setup timing becomes worse. Explain why and how you would handle the trade-off.
- After routing, setup slack becomes significantly worse compared with post-CTS timing. What would you investigate?
- A critical path has good cell delay but extremely high wire delay after routing. What physical changes could improve it?
- You have a setup violation caused mainly by a long interconnect. Would you upsize the driver, insert buffers, move cells, or change routing? How would you decide?
- You have a setup violation caused mainly by excessive cell delay. What optimization techniques would you consider?
- A path has a very small setup violation, but fixing it with cell upsizing creates congestion. What alternative approaches could you try?
- A path has a severe setup violation and very little physical space is available around the cells. How would you approach the problem?
- You have a hold violation on a very short data path. What is likely happening, and what would you do?
- A hold violation exists only in one corner but not in others. How would you analyze it?
- A path passes setup and hold at typical conditions but fails at a particular PVT corner. How would you debug it?
- A design passes timing at functional mode but fails timing in a scan/test mode. What could be different?
- After clock tree synthesis, one clock domain has excellent skew while another has very poor skew. What would you investigate?
- You discover that two registers belonging to the same clock domain are physically very far apart. How can this affect timing?
- A high-fanout reset net is causing transition and capacitance violations. How would you fix it?
- A high-fanout enable signal is causing timing violations. What physical optimization techniques could be used?



- You insert many buffers to fix a high-fanout net, but congestion increases significantly. How would you balance timing and congestion?
- Your design has acceptable timing but extremely high buffer count. What problems could this cause?
- A design has low utilization but very poor timing. What could be the reasons?
- A design has high utilization but excellent timing. Would you immediately reduce utilization? Why or why not?
- IR drop is very high in one localized region of the chip. How would you identify the root cause?
- You increase the number of power stripes, but IR drop is still high. What else would you investigate?
- An IR-drop hotspot is located near a large macro. What could be causing it?
- You have an EM violation on a power stripe. What physical changes would you consider?
- IR drop is clean during static analysis but fails during dynamic analysis. Why could this happen?
- A timing violation appears only when IR drop is considered. Explain how IR drop can affect timing.
- You have severe routing congestion after adding power straps. Why can power planning affect signal routing?
- You increase power-grid resources to improve IR drop, but routing congestion becomes worse. How would you find a balance?
- A design passes DRC before detailed routing but has thousands of DRC violations after routing. How would you approach debugging them?
- You have a large number of minimum-spacing violations concentrated around one macro. What could be the cause?
- You have many via-related DRC violations in a congested region. What possible solutions would you consider?
- Your design has antenna violations on long metal routes. Why do they occur, and how would you fix them?
- Adding antenna diodes fixes antenna violations but increases area and capacitance. What trade-off are you making?
- LVS reports that the layout and netlist do not match. What would you check first?
- LVS reports an extra device in the layout. What could be the possible causes?
- LVS reports a missing connection. How would you debug the issue?
- Your design has thousands of DRC violations after routing. Would you fix them individually? Explain your approach.
- A routing change fixes one DRC violation but creates several new violations. How would you handle such a situation?



- After routing, a critical net has excessive coupling capacitance. What can you do to reduce crosstalk?
- A victim net experiences a large delay change because of an adjacent aggressor net. What physical techniques can reduce this effect?
- A clock net is experiencing significant crosstalk. Why is this particularly concerning?
- You have a crosstalk-induced hold violation. What possible fixes would you consider?
- You have a crosstalk-induced setup violation. How would your approach differ from fixing a hold violation?
- Your design has high congestion because many critical nets are concentrated in one area. What placement or floorplanning changes could help?
- Moving a group of cells improves congestion but makes timing worse. How would you decide whether the movement is acceptable?
- A macro move improves timing but creates a new congestion hotspot. How would you evaluate the trade-off?
- You are given two possible floorplans: one has better timing but higher congestion, while the other has lower congestion but worse timing. Which would you choose and why?
- Your core utilization is only 60%, but routing congestion is still high. What could explain this seemingly contradictory situation?
- You have enough routing resources overall, but one metal layer is heavily congested. What would you investigate?
- Horizontal routing demand is much higher than vertical routing demand. What could cause this imbalance?
- After detailed routing, a large number of nets are detoured significantly. What does this indicate?
- A timing-critical net is heavily detoured by the router. How would you improve its route?
- A critical path contains several buffers inserted during optimization. How would you determine whether the buffering is actually helping?
- Your design has a large number of long nets after placement. What floorplanning or placement changes could reduce them?
- A critical path crosses several hierarchy boundaries and has long interconnects. How could hierarchy and physical placement affect timing?
- You have a large block with several input/output ports. How would you decide where to place the ports?
- Poor I/O placement is causing long paths between macros and boundary pins. How would you improve the floorplan?
- A macro has many connections on its left side, but most of the connected logic is placed on the right side. What problem might this create?



- You notice that many nets are crossing the same narrow channel between macros. What would you do?
- A placement has excellent timing but leaves very little whitespace around macros. What risks does this create later in the flow?
- You have large whitespace regions in the floorplan but severe congestion elsewhere. What does this tell you?
- A block has high density near the clock-tree roots. What problems might this create during CTS?
- A clock tree has excessive buffer count. What could be causing it?
- Clock latency is very high after CTS. What factors could contribute to it?
- Clock skew is within specification, but clock power is extremely high. What could be causing this?
- CTS meets skew targets but creates significant routing congestion. How would you optimize the clock tree?
- After CTS, a few endpoints have extremely late clock arrival. How would you debug those endpoints?
- You have setup violations on one clock path and hold violations on another path using the same clock. How would you analyze the two paths?
- A setup violation disappears after reducing clock uncertainty. Would you consider that a valid physical fix? Why or why not?
- A designer asks you to fix timing by modifying the SDC constraints. What would you verify before making any constraint changes?
- A false path constraint removes thousands of timing violations. How would you determine whether the constraint is actually valid?
- A multicycle constraint improves timing dramatically. What would you verify before accepting it?
- Timing reports show many unconstrained paths. What would you do before trusting the timing results?
- Your design has excellent WNS but poor TNS. What does this tell you about the timing violations?
- Your design has poor WNS but relatively small TNS. What does this indicate?
- After an ECO, timing improves but DRC violations increase. How would you handle the ECO?
- You need to fix a late-stage setup violation without changing the RTL. What ECO techniques could you consider?
- You need to fix a late-stage hold violation without affecting functional behavior. What options do you have?
- A timing ECO improves one path but degrades another path. How would you evaluate whether to keep the ECO?



- A design passes STA but fails DRC. Can the chip be considered signoff-ready? Why not?
- A design passes DRC but fails LVS. What does that tell you?
- A design passes STA, DRC, and LVS but still fails another signoff check. What other signoff areas might you investigate?
- You receive a block with high congestion, poor timing, and IR-drop violations simultaneously. What would you fix first, and why?
- You receive a block with good timing but poor IR drop and EM. How would you optimize it without destroying timing?
- You receive a block with good congestion and IR drop but severe setup violations. What would be your timing-closure strategy?

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